

Business Requirements Description

1. Purpose of the Document

This Business Requirements Description (BRD) defines the business needs, scope, and high-level requirements for the **Phishing Email Detection System**. It translates the Business Model Canvas into clear, actionable requirements that guide system design, development, and evaluation.

2. Business Objective

The primary objective of this project is to provide an **affordable, reliable, and intelligent phishing email detection solution** for organizations and individuals who lack access to enterprise-grade email security systems.

The system aims to:

- Reduce phishing-related risks and incidents
- Improve visibility and awareness of email-based threats
- Provide explainable and auditable detection results

3. Stakeholders

Primary Stakeholders

- Small and Medium Businesses (SMBs)
- Universities and educational institutions
- IT administrators and system security teams
- Individual advanced email users

Secondary Stakeholders

- Cybersecurity trainers and awareness providers
- Developers and system maintainers
- Hosting and infrastructure providers (if SaaS deployment is used)

4. Business Scope

In-Scope

- Detection of phishing emails using machine learning and rule-based analysis
- Classification output (Safe / Suspicious / Phishing)
- Confidence score and explainable detection reasons
- Email logging and audit trail
- Quarantine of suspicious or phishing emails
- Web-based dashboard for monitoring and management
- Self-hosted and optional SaaS deployment

Out-of-Scope

- Spam advertisement filtering
- Malware execution or sandboxing
- Antivirus file scanning engines
- Browser extensions
- Third-party threat intelligence APIs

5. Business Requirements

BR-1: Phishing Detection Capability

The system shall detect phishing emails by analyzing email content, headers, and embedded URLs using a hybrid approach combining machine learning and rule-based checks.

BR-2: Classification Output

The system shall classify emails into one of three categories:

- Safe
- Suspicious
- Phishing

BR-3: Explainability

The system shall provide a clear explanation for each decision, including:

- Key indicators that triggered detection
- A confidence score to support decision transparency

BR-4: Logging and Audit Trail

The system shall maintain logs of all analyzed emails, including classification results, timestamps, and indicators, to support auditing and incident review.

BR-5: Quarantine Management

The system shall quarantine emails classified as phishing or suspicious and allow administrators to review or release them.

BR-6: User Interface

The system shall provide a web-based dashboard that allows users to:

- Upload or view analyzed emails
- Monitor alerts and detection statistics
- Manage quarantined emails

BR-7: Deployment Flexibility

The system shall support:

- Self-hosted deployment for organizations
- Optional hosted (SaaS) deployment

BR-8: Extensibility

The system shall allow:

- Retraining of machine learning models
- Addition or modification of detection rules
- System updates without full redeployment

6. Non-Functional Requirements

NFR-1: Performance

- Email analysis should complete within seconds for normal workloads.

NFR-2: Scalability

- The system should support growth in email volume without significant performance degradation.

NFR-3: Security

- Secure access control for the dashboard
- Protection of stored logs and quarantined emails

NFR-4: Privacy

- Use of anonymized or redacted email data where possible
- Compliance with data protection best practices

NFR-5: Usability

- Simple and intuitive interface suitable for non-expert users

7. Assumptions and Constraints

Assumptions

- Users have basic technical knowledge for self-hosted deployment
- Public or anonymized datasets are used for model training

Constraints

- Limited budget and infrastructure for large-scale SaaS deployment
- API-free operation to reduce dependency on external services

8. Success Criteria

The project will be considered successful if:

- The system achieves high phishing detection accuracy
- False positives are kept within acceptable limits

- Users can easily interpret detection results
- The system is deployable and maintainable with minimal overhead

9. Business Risks

- Model degradation due to evolving phishing techniques
- Dataset bias affecting detection accuracy
- User trust issues caused by false positives

Mitigation:

- Continuous retraining and rule updates
- Feedback loop for detection improvement

10. Conclusion

This BRD defines a clear business-driven foundation for the Phishing Email Detection System. By aligning technical capabilities with real-world customer needs, the system delivers practical, scalable, and explainable email security suitable for academic, organizational, and commercial use.